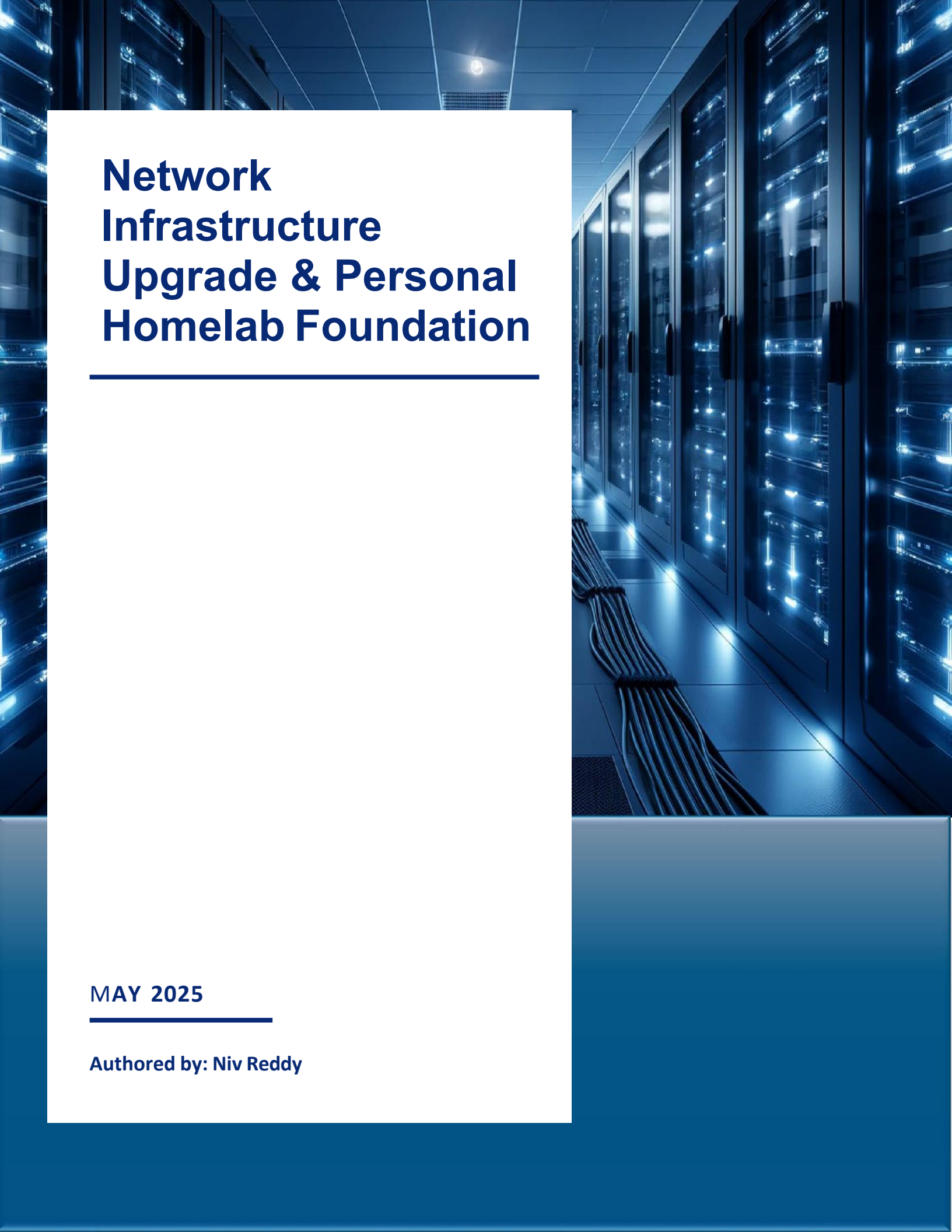




Network Infrastructure Upgrade & Personal Homelab Foundation

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Authored by: Niv Reddy

Introduction

Motivation, Goals, and Project Scope

This project began with a simple goal: connecting my Dell OptiPlex to Ethernet so I could install Ubuntu and begin learning about server management. That small goal led to a full redesign of my home network. I researched mesh Wi-Fi, backhaul setups, and proper network segmentation. I eventually converted our ISP-provided Xfinity modem/router combo into bridge mode, made Eero the main router, and rewired my network panel to support a wired mesh system.

My goal was to gain real experience with IT hardware, home networking, and security practices. I wanted control and visibility over my own network—something I never had with the default ISP setup.

Equipment Used

Network Equipment and Installation Tools

Component	Description
Xfinity Gateway	ISP modem/router combo, now in bridge mode
Eero Pro 6 (x2)	Original Lennar-provided unit acting as main router
Eero 6+ (x3)	Added units placed throughout house w/ wired backhaul
TP-Link 8-Port Switch	Unmanaged switch for Ethernet distribution
Crimping Kit	Terminated Ethernet wall ports with RJ45 connectors
Cable Tester	Verified pin order and continuity
Cable Management Tools	Adhesive mounts, Velcro straps, panel cleanup
MoCA 2.5 Adapter (x2)	Converts Ethernet to coaxial and vice versa to enable wired connectivity via existing coax lines. Used to deliver Ethernet from the central panel to a bedroom using in-wall coax.

Network Setup & Layout Overview

Wiring Layout, Configuration Flow, and Device Placement

After evaluating my existing setup, I saw potential to upgrade from a basic Xfinity modem/router combo and a single Eero to a fully wired mesh network. My home came pre-wired with Ethernet runs, and once I discovered these access points, it became clear I could leverage them for a more efficient and scalable setup.

I started by converting the Xfinity gateway into bridge mode, disabling its routing features and allowing my Eero Pro 6 to function as the main router. This Eero unit was originally placed inside the internet panel, but I moved it just outside the panel to avoid signal interference. From there, I connected it directly to an unmanaged 8-port TP-Link switch that I mounted inside the panel.

Using a crimping kit and tester, I terminated all pre-run Ethernet lines coming from different rooms in the house. Once verified, these lines were plugged into the switch, enabling wired backhaul to each new Eero 6+ node placed in the upstairs loft, master bedroom, and downstairs living room.

This configuration allowed every Eero to operate on Ethernet instead of Wi-Fi, significantly improving speed and reliability. All cable runs were organized using adhesive mounts and Velcro straps for a clean and maintainable setup.

This wired mesh layout forms the foundation for expanding my network into a full home lab in the future.

To extend Ethernet access to my bedroom without running new cable, I installed a pair of MoCA 2.5 adapters. This allowed me to use existing coaxial wiring to transmit a wired connection between the central network panel and my room. One adapter is connected to the TP-Link switch, while the other is placed in the bedroom and feeds into a local device or additional switch. This significantly improved connection stability for devices in that area without relying on wireless backhaul.

Network Topology Diagram

Visual Representation of the Final Wired Mesh Layout

The following diagram illustrates the finalized network topology after the full upgrade. It shows how each component is connected—from the Xfinity modem (in bridge mode) through the main Eero gateway, switch, and individual wired-backhaul Eero nodes placed throughout the house.

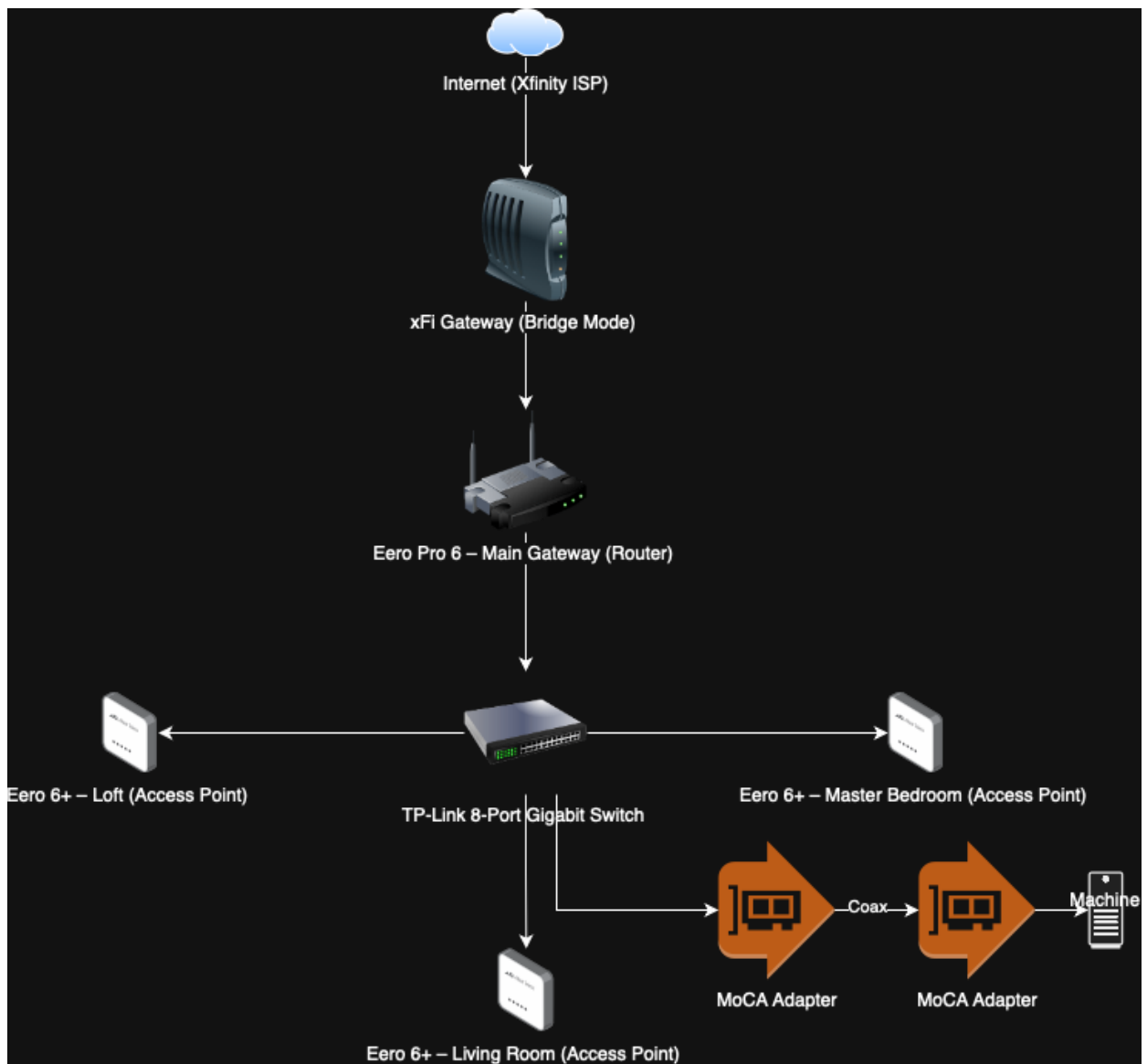


Figure 1: Home Network Topology with Wired Backhaul

Configuration Settings & Security

Router Setup, DNS Customization, and Network Hardening

After finalizing the physical setup, I turned my focus to optimizing the network configuration and securing the infrastructure. The first step was enabling bridge mode on the Xfinity xFi Gateway, effectively disabling its router functionality so the Eero Pro 6 could take full control as the primary router. This ensured there was no double NAT, which can cause interference with port forwarding, VPNs, and overall stability.

Next, I configured the Eero network by creating a new SSID and connecting all my personal and IoT devices. I renamed each device within the Eero app to easily identify them and monitor bandwidth usage. This visibility gave me granular control over who and what was active on the network.

To further improve reliability and privacy, I changed the DNS settings from automatic to Cloudflare DNS (1.1.1.1 and 1.0.0.1). This switch provided faster resolution times and enhanced privacy by preventing my ISP from tracking DNS queries.

I also configured a guest network through the Eero app to isolate visitors from the main LAN. This offered a clean separation without requiring manual VLAN setups.

From a security standpoint, I ensured WPA3 encryption was enabled across all access points. I verified that UPnP and Smart Queue Management (SQM) were disabled or left off, based on best practices for a basic home network that didn't require advanced QoS. Notifications were also enabled to alert me when new devices attempted to connect.

These configuration and security enhancements laid the groundwork for future home lab experiments and ensured the network was optimized and protected.

Challenges & Lessons Learned

Hands-On Experience, Trial and Error, and Growth Through Iteration

One of the first challenges I faced was dealing with the unknown wiring in the internet panel. Although the house was pre-wired for Ethernet, the cables were uncrimped and unlabeled, leaving me unsure which ports connected to which rooms. I had to learn how to properly crimp RJ45 connectors using the T568B standard and test continuity with a cable tester to ensure successful connections. I had to redo a few cables due to miswiring and failed tests, which taught me the importance of pin order and double-checking crimps.

Another major learning moment came with understanding the role of each device. Initially, I didn't realize the conflict that occurred from having two active routers (the Xfinity gateway and the Eero Pro 6). This caused confusion with double NAT and inconsistent performance. Researching the concept of bridge mode clarified that I needed to disable routing on the ISP-provided modem/router combo so my Eero could operate independently and handle all traffic.

Physically managing the setup also posed challenges. The internet panel was cluttered and tight on space, making it difficult to mount and wire everything cleanly. I overcame this by using adhesive mounts, Velcro straps, and shorter patch cables to achieve better organization and airflow.

Finally, while testing the network, I noticed that speeds weren't drastically higher, but stability and device visibility had significantly improved. I also learned that not all improvements are about raw speed—sometimes, network reliability, manageability, and learning from the process are more valuable.

This project taught me that hands-on experience—even in a home setting—can offer valuable insight into network design, hardware configuration, and troubleshooting. These lessons laid a strong foundation for future home lab expansion.

Future Plans & Homelab Roadmap

Expanding Infrastructure, Learning Enterprise Concepts, and Applying Them at Home

With the home network infrastructure now upgraded and stable, I plan to expand this foundation into a full home lab environment focused on real-world IT and networking concepts. My next step is to replace the current Xfinity gateway in bridge mode with a dedicated standalone modem, such as the Arris S33. This change will eliminate any residual firmware or software conflicts caused by the existing combo device, ensuring a cleaner handoff to the Eero system.

On the server side, I plan to repurpose my Dell OptiPlex as a local server running Ubuntu. It will initially serve as a sandbox environment for testing basic server functionality, running services like SSH, Samba, and potentially hosting a private media server. I also intend to install Proxmox or VirtualBox to experiment with virtualized environments.

One of my key goals is to create an Active Directory (AD) lab. This will allow me to simulate user/group management, domain setup, DNS configuration, and group policy deployment. It's a hands-on way to understand what real enterprise network environments require, while building administrative and troubleshooting skills.

Additionally, I want to implement network-wide DNS filtering using Pi-hole to block ads and improve security. In the longer term, I plan to create separate VLANs to segment my IoT devices from the main network and experiment with firewall rules, VPN access, and monitoring tools like Grafana and Prometheus.

Finally, I also plan to set up a NAS to store media files, project backups, and photo/video archives. As a photographer and videographer, integrating my creative workflow with my homelab setup will make it both functional and personally useful.